2 – August Focus Group – FG1

Interviewer

So, kind of as a bit of an ice breaker and for people to introduce themselves, we're just kind of going to go through everyone. And it kind of be an opportunity for you. Oh, well, what I want everyone to do is kind of say what type one diabetes technology they're using at the moment, in terms of like the pump or CGMs they have, what smartwatch they chose as part of this or what they're using. And then also one bit of diabetes technology you've used that you dislike more or maybe hated even. Um. So I'll go first. So. Um. Well, at the moment, I've just swapped over to the t:slim, t:slim X2, so that's been really good to see. I've swapped to the G6 sensors as well. I'm having to self-fund those at the moment, which is a bit of a pain, but I'm hoping with all the stuff coming out that that will shift in the next few years. I have a Fitbit Sense actually, I used, I kind of got it as part of the PhD to try and get a feel for smartwatches and what they might be used for. So, I've kind of been using that and have an idea if that works. And then, my bit of technology I have, that I'm less a fan of is, I used to be on the Accu-Chek Spirit Combo and the blood glucose reader, which I have here, started really good because it's kind of Bluetooth together and you think that would be kind of very useful to have. But, unfortunately, every time you try and use it um, it connects the pump, even if you don't want to use it. So, it just means there's kind of that 20 seconds delay that's trying to connect sometimes doesn't even manage it, and it just kind of extends that process out and makes it that little bit harder. So yeah, so if you just kind of do the same and go through, just get a feel and I'll kind of pick out a random, but [P06] do you mind starting.

P06

Hello, can you hear me? Yeah, yeah. OK, great. So I'm [P06]. I'm using a FreeStyle Libre 2 as my CGM. Um. I'm on the Omnipod, basic one and I've been using that for quite a while because on the whole I think it's really good. One thing I really don't like about it though is the PDM. I know there's a new updated version which is much better, but it felt like it took them a while to come out with that and the kind of UI on it is awful. Um. So yeah, if it now they've come out with a new one, I think be quite interesting to try that and see how it compare. Um. And then the Fitbit I got was the Luxe, which is kind of the basic option. Erm. And I've never used a smartwatch before, so it's been quite interesting kind of keeping that on your wrist the whole time and having a bit more extra data to hand.

Interviewer

There any bits of technology in kind of type one diabetes you've not enjoyed so much or?

P06

Well, actually I used to use the same er pump, you had there and one thing I found about that is while I was running around, especially at school, I remember catching it on like door handles and the whole cannula getting ripped out, which I really did not enjoy.

Interviewer

Yeah, I guess that makes sense, since the swap to the Omnipod that fits, yeah. I think I've definitely experienced that before as well, it's not particularly pleasant. Do you mind um picking someone else then to go next?

P06

Uh, yeah. The first one I've got up is [P19].

P19

Yes. Hi, I'm [P19]. I'm currently using the Dexcom G6 and the Tandem t:slim X2. I've been on G6 for quite a while now. And these Tandem slim I got some time last year. It's good. I I mean, I'm a little bit annoyed with it today because it it was decided to shut itself off randomly this morning. Um. But yeah, it's been very, very good. My diabetes tech that I don't like is basically anything from Medtronic. I was with them, it's my first ever insulin pump, I've had two of them, it's over 10 years of Medtronic and I just absolutely hate their entire way of working. Particularly their customer service as well. I don't know if anyone here has spoken to the Americans, they are just absolutely your awful. But yeah.

Interviewer

Yeah. I'm sorry that. Yeah, that does sound like a bit of a pain. What? What was the smartwatch we're using as part of this?

P19

Yeah. So I'm using my own smartwatch, so I've got the Apple Watch Series 7, which I've had for about a year and a half or so I think, which is the third generation of Apple Watch that I've had.

Interviewer

Right. OK. And do you mind picking out someone else to go next?

P19

Yeah, [P18].

P18

Hello. Um. I'm [P18]. I am on a Medtronic 780G. Just stand alone at the minute, I've not got the sensors that make it a closed-loop, which is a separate battle. What's the second question? Some of the tech I don't like. It is a bit of a nod to the Accu-Chek because I was on that for about 10 years.

The combo, first of all, the speed, but also just the clunkiness of it. You know, in a lot of ways, it's not the most discrete pump I had. The other thing I wanted to sort of mention is my sensors at the minute I'm not very impressed with. It's the Dexcom One, which is what the NHS are paying paying out for. Up to now I was self-funding though, so I was doing the Libre 2 and the Dexcom G6 in separate sort of tandems and they were really good. The G6, it's much better than the One. Simply put, I keep getting signal loss. I've even got it now, signal loss. So, I can't actually see what my blood sugars are. And it actually got really serious last week because because of the signal loss, it didn't alert my phone and I actually had my first ever diabetic seizure last week where I had to have like, paramedic assistance and everything. And they were like, well, why are your sensors not working. So, I've been complaining about that. And then separately to that, the smartwatch I'm using, was the Sense I believe. But I've reverted back to my Apple Watch now.

Interviewer

No really I'm I'm sorry to hear about that. That sounds like, yeah, quite scary.

P18

Is is crazy. I've never had one before. But that might be important for you to hear that it's a wearable tech issue.

Interviewer

Yeah. No, I and yeah, hopefully something they should have sorted or would have sorted. By now, yeah, I do. You mind picking out someone else to introduce themselves?

P18

Yeah, [P03].

P03

Yeah. So, I'm on the Tandem t:slim X2 and the Dexcom G6 combined to make it a closed-loop system. I'm really enjoying those. One of the benefits of the Dexcom is being able to share it with my friends. So, I was at a festival a couple of weeks ago and just being able to have my friends all know what my blood sugars were was really helpful. If I was like going low and they could be like “you need a snack, come on, let's go get a snack”. A piece of technology I've had in the past that I didn't like was the Accu-Chek, also, the combo. Mine ended up with a manufacturing issue and I ended up in hospital because I was in DKA because I was just not getting insulin because the pump failed.

Interviewer

Yeah, this is a bit of a, Accu-Chek are taking a bit of a beating today, I think, but.

P03

Yeah, yeah. Do you always pick someone to go next?

Interviewer

Yeah. What? Um, What smartwatch are you using actually?

P03

Ohh sorry yeah, I'm using the Fitbit Sense.

Interviewer

Fitbit Sense, brilliant. Yeah. Yeah. And if you could pick someone else to.

P03

Err, [P24].

P24

Yeah. Hi. I'm on the Dexcom G6 and the t:slim X2 as well. I've been on the Dexcom, I've been on the G6 since it came out. I was on the G5 and the G4 before that, and it recently just got funded by the NHS. Um, when I went on to the t:slim in January, they started funding my Dexcom with that. I obviously I like the Dexcom, I've been it been with it for ages. But, the t:slim has been working really well for me as well. And I I probably agree with [P19] that like the the main diabetes technology that I absolutely hate is anything Medtronic. Like even just changing over to the t:slim, the cannulas never fail. Like I've had maybe one cannula fail on me since I started the t:slim, whereas Medtronic, it will be like maybe once in one in every three sensors. And I've ended up in hospital because I was in Germany and ended up in hospital because my cannula had a hook in it. And it's just like it was such a constant issue. Not to mention how awful the Medtronic pump was having moved over from Animas. But yeah, I just hate Medtronic. And the smartwatch I'm using is the Luxe, the Fitbit Luxe.

Interviewer

Great. Thanks for that. I think that just leaves.

P24

I think, yeah, just [P21].

Interviewer

Oh, [P21], can you hear me alright? We can't hear anything coming through, if that's. Yeah, I can't hear anything. I don't know if any of the others it's coming through or.

P21

Is that?

Interviewer

Oh, oh, I heard you briefly and then you yeah.

P21

Ohh hello.

Interviewer

Perfect. Thanks very much.

P21

OK. Hi sorry sorry about that. So I am using the Omnipod Dash and the Libre 2 at the moment, which I'm finding is working pretty well especially. I mean, there's there's been pros and cons of of the update, but since they've had the the update where you no longer need to scan on the Libre 2, I've found that generally that's been pretty useful, being able to see things immediately. Although, there has been a few issues with signal loss. I don't know if anyone else on on the Libre has found that. Um, but hopefully just sort of teething issues and and that'll be sort of eradicated. Um, I'm using the Fitbit Luxe as my smartwatch, which I'm really enjoying so far. It's one of the more more basic models, but I think it's worked really well for for me. And I'd say a piece of tech I dislike is another vote for Medtronic. I was on the Mio I think I think. Until for about 8 years. Yeah. I think for me a a non-wired pump is a lot better. I can't remember who it was, sorry that mentioned about sort of getting things stuck on on door handles and things like that. I think for me like the the wireless tubeless element of the Omnipod is definitely a big plus. Um. So, yeah.

Interviewer

Yeah. No, that's great. No thanks everyone for doing that. And then. So we'll kind of like proceed into the general questions then. They will kind of start off just open, so if you've got something, say please dive in. And also if you got something to follow on from someone else, that would be great. If you agree, disagree, whatever that would be really good to see. But I guess. So, first off, it's just kind of again as we did last time, kind of look over the last month and see how you found using the

wearable. And I know people are on lots of different ones and people are using it different lengths of time, but I guess first of all, is there. What's the feature that you think you find yourself most commonly using or have been for the last month?

P06

I can go first. Recently I've been using the step counter, just because I've just been having discussions people about how you should be hitting 10,000 steps a day. So now I can actually see how many steps I'm doing, it's kind of interesting to see that what I thought 10,000 steps actually is in a day. Because people used to say that and I'd have no idea. So, I found that quite interesting.

Interviewer

Yeah. Does anyone else use this kind of 10,000 step goal at all or?

P18

I just used like the activity tracker as a whole for the Apple Watch. It sort of summarises it all into one app, so that's move, so that's active calories, that's standing hours, as well as actual activity. And then further down, it's got the the steps and the distance travelled and floors climbed. Yeah, that's probably my most used app.

P21

I'm similar to [P06] in that I've been using the step counter specifically because you can have it sort of on your home screen on the Fitbit. So sometimes I'm walking around I like to see what I was on before, what I'm up to now, how many steps my walk has sort of added on. And I do use that similar 10,000 step indicator as well.

P24

I found that I would like previously I would just look at my step count on my phone and go into the health app for it, but it's been quite nice being able to have it much more easy accessible. And I kind of get a little bit of a buzz when it it goes off, even if I'm just like walking around my room and I happen to hit it while I'm walking around my room, it's quite nice to know that I've hit that goal even though I'm not that fussed about it from a from like a a health perspective.

Interviewer

Yeah. Does anyone else get that kind of similar buzz when they kind of hit that 10,000 point or something equivalent that they might be aiming towards?

P06

Yeah, I've had it. The one thing that seems to happen a few times, though, is I'm sitting still, and then it goes off, which isn't super encouraging. I was wondering, [P24], how does that, how do you think it compares to your like phone recording it?

P24

Probably much more accurate, because the only time I take my Fitbit off is to charge it and I just do that while I'm in the shower, so I'm not doing any steps in the shower anyway. Whereas my phone obviously I not, there's no guaranteed that it'll be on me all the time. I feel like just having it on my wrist is a much more accurate way of recording where I'm actually moving. But then again, I haven't been comparing the numbers directly to what my phones been recording.

P06

Fair enough.

P03

I have noticed sometimes I'll wake up in the morning, like I've slept all night, but I've done 110 sleep steps in my sleep and I'm like, “starting the day off right? I just started sleepwalking”. But I think I use. I do use the step counter quite a lot, but I also use the sleep tracking mode to see how well I actually slept in accordance to how I feel.

P24

Yeah, I love the sleep tracker.

P06

Yeah, I gotta agree with that.

P03

Yeah, because I'm quite a bad sleeper. And so I'll use like the data of, I feel a bit groggy this morning, it looks like I didn't sleep as well, I’ll try and get myself to bed earlier.

P24

Yeah, I recently had my Fitbit app suggested like a sleep course for me to take this morning. Which you know, I I was intrigued, so I clicked on it to see what it's all about. And it basically just has set me like a goal that I can reach based on how much sleep it thinks is appropriate for me, based on like all the data that it gets. And then also it's like done a little checklist of things that I can do to help my sleep help my sleep be better so like when I.

Interviewer

Oh, sorry [P24], I think you might have caught that briefly. Yeah, I think, uh, it sounds like, uh, maybe it's the connection issues. But yeah, no, I think that's interesting what people are saying about or [P03], you mentioned that you kind of made a change as a result of seeing that kind of sleep stuff. Has anyone else kind of made adjustments after seeing that information on the smartwatch of some kind?

P21

Yeah, definitely. I mean on the same same note of the tracker, I'm not gonna lie, I was shocked when I looked at the data from it, cause I thought I was getting 7-8 hours sleep. And then sometimes when I look at the data, it would actually be a lot less or I was waking up at times in the night, or sort of restless. So, I've definitely been trying to implement some changes and and go to bed earlier and and make sure that I get a full full night's sleep.

P06

Yeah, me third. I think I've definitely been way more conscious about trying to get the most hours of sleep as I can. The one thing I would say that's got me slightly unconvinced by it, is the recorded like 3 hours sleep, but the score is still like 68 and it says good. So, that's made me slightly wary of like, what is it actually reading and how is it scoring it. But on the whole, it is quite interesting to look at, and see the breakdown of different types of sleep you're getting.

Interviewer

Yeah, I think that's interesting, what you're saying about kind of, yeah, the sleep score being there and kind of not, not really know what it it means necessarily. Is that something you kind of want to have a better understanding of or?

P06

Yeah, I think if it, I mean to be fair, I haven't really tried looking into it at all, but it wasn't immediately obvious how they came up with the score. Similar to the stress management, it just slaps the score on there, but if I click on it, I couldn't really seem to understand what it was saying. But, then again, I wasn't like digging deep to try and, it was a very much a brief click on. It was quite unclear about what the score. Presumably higher meant good, so that's what I was basing it off of. Uh, yeah.

Interviewer

Yeah, no, definitely. Has anyone else experienced that? There are bits that are kind of been a bit harder to understand or interpret exactly what they mean?

P24

Yeah, definitely. Some of the scores just aren't very kind of very self-explanatory.

Interviewer

Yeah, I guess. And I, I guess most of those comments around kind of the Fitbit, I guess maybe [P19], if you're on the Apple Watch, how have you found that? Is that similar or different?

P19

I mean, I'll. I'll be totally honest. I I don't use the the health tracking features basically at all on my Apple Watch really. A little bit of waste, I've just never really gotten into them. So, I'll occasionally glance at things like how many steps I've done in a day or whatever, but I mostly rely on it it tracking what activity I do and kind of logging it. But the main thing I use a smartwatch for, what I've always used smartwatch for is for like the notifications, the phone calls the text, that kind of thing. And now for the diabetes management as well.

Interviewer

Right. No, totally. Do other people use the like most like notifications texts or calls, etcetera?

P24

Yeah, it's, it's. I've definitely missed fewer phone calls because I actually notice it ringing.

P06

What I found quite useful actually is um I've been in meetings at work and I feel my phone vibrate and I feel my watch vibrate and I normally if they both vibrate it means it's a low glucose alarm or something. It's quite distinct vibration, which I hadn't realised I'd realised but. UM. It's kind of a bit more subtle to just quickly look at my watch to see what my control is doing. I don't know if someone on like a a more advanced mode actually you can actually get the live graph on there. Because, that's something I've found, is that like it says I'm going hypo, but I don't know how hypo I am going. Which, would be quite useful in the middle of a meeting, because if it's like I've just dipped in briefly, I don't have to like, I've gotta go quickly.

P24

I'm wondering, like I haven't managed to get my watch to also vibrate where my bloods are.

P06

I would say it's it's inconsistent. Like sometimes it does vibrate, sometimes it doesn't. But I think I was I was able to set it up through one of the settings. I can't remember if that was.

P24

OK, because all my phone does, all my watch does is vibrate when I get a phone call. I don't it doesn't connect in any other way.

P06

I yeah, definitely in the setting somewhere. I can't remember if it was on my phone or through Fitbit, but there was a way of setting it up.

Interviewer

Yeah, it's like that from the Libre app itself, the notification kind of coming through that on the phone or is it something different?

P06

Yeah

Interviewer

Right, I see. Does anyone else use the watch to get their readings through? Or, yeah, what's the kind of split on that?

P18

I probably, I'll notice the notification and then look straight at my watch and then it will say, you know the Dexcom sensors got like their like high alert, going high, but then that's the sort of extent of the notifications. So yeah, I just do that. That's why I use it for just the the notification side of it. But also, I suppose because I'm quite an Apple girl, I've it like hand off as well as like other Apple and sort of Airplay and things like that.

Interviewer

Yeah, I guess there's there's different sensors that allow you to kind of see the readings that come up on it. I remember people saying that before. I know on the Apple Watch you can get the Dexcom sensor through, but also there's the Glance app and some of the Fitbit. Has anyone been using that at all? I mean it is that it's something you wish was there. If you don't currently have the readings coming through.

P06

Sorry, what is the Glance app?

Interviewer

Well, has anyone used the Glance up actually? Or tried it out?

P03

I tried to get it to work. I don't think I've actually managed to set it up yet. It's just a way to get your Dexcom graph onto your Fitbit. I don't know if it connects to the Libre or not, but I know it can, it can connect to Dexcom if you manage to get it set up.

P06

Cool. Thanks.

Interviewer

Like I guess maybe on a different track then and thinking back about the smartwatch more is, has anybody tried out anything new in the last month compared to that first month that they hadn't tested before? I don't, I don't know. Like. Yeah, [P21], say, have you test that anything new or is it just kind of the same stuff as the previous month?

P21

In terms of the different applications on the watch? Um. I wouldn't say. I think I've sort of been using the same same stuff since the last time we spoke, I think I've I've delved sort of deeper into the settings of, like the sleep analysis, like looking at the different stages of the sleep cycle like REM and and deep sleep and things like that. And using sort of the period tracker and as well as you know the the activity things.

Interviewer

Yeah. And the kind of amount you're using it do. You think that's changed much or?

P21

I think it's probably consistent. I mean, I'm definitely getting more used to the different features and how to use different settings, yeah.

Interviewer

Yeah, has. Has anyone else kind of, do they think it's changed and kind of that second month of using it? In terms of how much they're using, that is, the frequency of interacting with it.

P24

I think I've probably um like I'm not looking at the data quite as much, because in that first month it was like really exciting every single night like, “oh, what's my sleep score? How many minutes a week did I have?” And and I'm less I'm I'm less bothered about looking at it every day now, but it has been really nice to be able to see my averages that obviously weren't quite as built just after that first month and compare how I've been doing.

Interviewer

Right. Interesting. And have you noticed much change then in those averages over that time?

P24

Yeah, I've. I think my like across the board, I've been doing like. My my blood, I know that my bloods have got a bit better, which has coincided with my like, for example my sleep getting a bit better. And I'm not sure, if that's just because I'm more aware of it or because one of them improving has improved the other. But you know, I have noticed an improvement, whether it's a permanent one, I have no idea.

Interviewer

Yeah. And I guess that will be interesting to see, over the next few months how that changes. Yeah, has anyone else got anything to comment on that?

P06

I'd say that I've probably started looking at it a bit more. I don't know what the first month I was kind of a bit sceptical that it wasn't gonna show me much, but I've kind of looked at my heart rate a bit more to kind of see if there I can think of any correlation between when my heart rate's been higher or my sugars doing stuff. And I've not like compared the numbers, but off the top of my head it's like I can see my heart rates jumped when I've been like, nervous or like anxious or something. And then it's like my sugars kind of seemed to stay the same in that point, so. Yeah, I'm kind of starting to notice less of a correlation than I thought there would be. When I first started I was kind of expecting there to be a bit more of an obvious correlation between them. But yeah, haven't seen much.

Interviewer

Yeah. Has anyone else noticed links between heart rate or anything else in the watch and then changes in the blood glucose levels? Is that, is that something anyone's looked for or been considering or it's just kind of been there?

P21

It's not something that I've looked for particularly, but I have noticed in certain situations where there's sort of high adrenaline, my blood sugars do tend to spike and alongside my heart rate. So, for example, like I've done a few like running races and normally I try and be about sort of 9 or 10 mmol/L before and I've noticed just before sort of going to the start line, my heart rate will go up and my blood sugars will spike up as well. So, I think for me, there's definitely a correlation between the two.

Interviewer

No, that's really interesting and I guess, yeah, is there anything else you kind of noticed and? I mean it seems like some people commenting on commenting on the stress element and the effect on heart rate, do you see other links between heart rate and changes when you're doing activity?

P21

I suppose, generally speaking, higher heart rate is sort of when you're exercising, which then normally low as low as glucose. But for me I find that it's normally afterwards, after the exercise, that my blood sugars start to come down, rather than during or or when the heart rate is actually higher.

Interviewer

Right. Interesting. I guess thinking more about the stress then is how consistent is the increase in heart rate that you've both seen?

P21

I think I think pretty consistent. If it's sort of just talking about the stress element, I know even at sort of work, if I sort of get a big task for me or something like that, I do find that I. It it is likely to make my blood sugars rise as well.

Interviewer

Right. Is that the same for you [P06], have you. How consistent is that increase in heart rate when you get stressed or?

P06

Um. The amount, it seems to spike by isn't that consistent? I think definitely depends on my mood. If I'm feeling like very tired and I'm definitely a bit more like anxious and it causes a bigger spike I think. Also, depending on how well I slept, yeah. So, I think I've noticed a correlation actually saying that between how well I've slept and how much my heart rate seems to spike I think.

Interviewer

Yeah. And I guess that kind of builds on to what we kind of, I've spoken with all of you about in the previous interview in terms of then can that data or information you're seeing on the smartwatch being used in a closed-loop system of some kind. You've mentioned a couple of things there then like stress and sleep and activity. Do you think there's anything else that could be involved in?

P24

I mean, personally I've I've found that like, even just if my bloods are high, my resting heart rate seems to be about like 10 or 20 higher than it usually would be. Um. And I don't know if that's cause I get anxious when my bloods are high, or if it is like more physiological, just my body responding. But um it seems like there's a relation there, relationship.

Interviewer

Right. I see. That's really interesting. Yeah. So the the other way around, more kind of your yeah. Your blood sugars then impacting what information would come up on the smartwatch.

P24

Yeah.

Interviewer

Has anyone else noticed something similar? [P19], have you ever seen trends between what's on the watch and?

P19

Very kind of anecdotally. I think generally when my heart rate goes up, it's generally when my blood sugar goes down, particularly in the mornings, if I wake up hypo, I'll find, you know, my heart rate's kind of pounding and everything. And then if I look at the associated kind of graph on the watch or whatever, you can kind of see it increase in some relation to the blood sugar decreasing. But, it's not amazingly consistent either, but it is it's something that happens I think when I get some more serious hypos than kind of pretty low key ones.

P03

I think I'd shared that experience with [P19] as well.

Interviewer

Yeah, yeah, I guess that it's kind of. Or yeah, kind of thinking about the system where it's linked in. What? What do you think it's going to help the most with or? What do you think it won't help with? Does anyone feel particularly strongly about there being a particular idea it would help with?

P24

I think it would be quite effective in responding to your heart rate in terms of exercise. Because, even just when I'm stressed I I can compare my heart rate when I've been stressed to when I've been exercising and they're quite different like. Even though they're both increases when I'm exercising, it tends to be for a lot longer and it peaks higher, so I feel like in a closed-loop system that could be used to inform like a reduction in the amount of insulin I have or whatever. But I think there would be perhaps less application for those short spikes that I get from stress.

P06

I'd definitely agree with that. That's kind of exactly what I was thinking. I currently don't have a closed-loop system so I think like while I'm sleeping, I think that'd be quite a good place that it could help with. Where it's like it's fairly consistent, but I I don't know if the closed-loop can do that already, just based on what your sugars are doing. I wonder if it could be more accurate with watch input maybe. Sorry go on.

P24

Yeah, sorry, I was just going to say that the closed-loop system that I used, the t:slim one, although it does have like it will respond to whether your bloods are high or low, it has different settings based on whether you're asleep, exercising, or just normal. So, I think one thing I'd mentioned to [Interviewer] in a previous interview was like it could potentially use your heart rate from when it can tell that you're sleeping, or when it can tell that you're exercising to activate one of those modes. That was the idea.

P06

Yeah, no, sounds like pretty good idea.

Interviewer

Yeah, [P18], what do you think about that?

P18

I I don't really notice my heart heart rate when it comes to physical exercise or in relation to bloods at all, really. So, I guess now that I've had this conversation that I might have a look, but I think that's definitely a really good idea. Because then it's that consciousness, but also try not be too blasé and rely on the tech, but if it's there you will use it, won't you.

Interviewer

Yeah, I think that's really interesting, what you're saying about the the reliance on the tech. Does anyone else have a have concern that they're too reliant on their technology?

P24

I was really, I was really worried going into the t:slim because it's such a change from a manual, manually operated pump and I was quite terrified. Like, oh, "what if it kills me off? What if it, like, goes completely wrong?" But you know, I just being able to let go and give it that control, it's been great. Because, kind of like you don't notice that it has more control because you're just not having as many loads and highs, so it's almost just like you don't even realise you're giving up that control apart from in the first couple of weeks. I feel like the same thing could apply to if it was a closed-loop system with your watch. If it went well, then you'd you'd forget that it was going well, if you know what I mean.

P03

Yeah, I can agree with that. When I first was put on the t:slim, there were times I was almost fighting with my pump because I was like, "I don't want to give up all this control, I don't trust it". And so, I'd end up going like super high and then super low because I was overriding what the pumps are and it was fighting back with me. But I did learn to.

P24

I did exactly the same.

P03

Yeah, I might. I I just definitely fought with the pump and lost. But now I'm more trusting of it. It's working really well for me.

P19

Yeah, I think I'm gonna echo what the other t:slim users are saying. My first even 3 months or so on a t:slim, it was weird having it intervene, but I had to just kind of trust it. Especially because like the Dexcom sensors have always work really well for me, giving really accurate data. So, but it was just so weird after being in control my diabetes for like you know, so long, 10 years, whatever like just giving it over to this algorithm. But then when I did, I realised that actually it's it's a lot for the better. Yeah,

but like this morning for me was a bit of a wake-up check cause my I was doing a pump upload for this and it literally my pump switched off completely and wasn't turning back on again. And I rang Air Liquide and everything and I was kind of in a bit of a panic. And I was like, “oh, God, I'm completely reliant on this”, cause I don't even think I have insulin injection pens like in the house. Because the backup I think I have an old Medtronic pump, so I'm definitely a little bit overly reliant on the on the technology I’d say.

P24

And they're similar, but less severe experience, that's more of just a funny, funny story really, as my pump ran out in public and I had to go to Curry’s and ask them at the till if I could charge my pump and just was standing by the till plugged into their system. And it was like, huh, I really like I can't get home in time, I need to find somewhere to charge my pump or I just die, essentially.

P06

I had an experience using the Libre 2 recently. They updated their app to give you the automatic reading so you didn't have to scan it. And then yeah, one day woke up and it just wasn't opening. The app was just crashing instantly, and I figured it'd be fine. I just told myself it'd be fine, but then like 2 hours later it still wasn't working, I really did start to panic and had to go find my older pen clicker to prick myself to give blood. So yeah, that was a bit of a shock having to go back to drawing blood.

P21

Yeah, [P06], I had the same thing with the the Libre update. I think I saw on some of the forums that it was sort of a worldwide issue where you had to sort of delete the app and then they're taken off the App Store. You had to like go and find an old download and find and get it back that way. But, I think that's sort of definitely showed me that we are quite reliant on the tech sort of same going back to finger pricks and you kind of feel like you get a much smaller snapshot doing 4-5 finger picks a day. Rather than, you know, the sort of unlimited data that that CGM or Libre can can give you.

Interviewer

Yeah, definitely. I think it would be interesting to hear from you as well. I guess hearing the other people who are talking about their experiences on the t:slim and closed-loops. Umm. Is it something you'd like to move on to? Or how do you kind of feel hearing about that kind of control element?

P18

Who was the question for sorry?

Interviewer

Oh, sorry for [P21]. I think just as she was saying. But [P18], if you have something to say.

P18

Well, I was just thinking the amount of positive um stories I'm hearing about it I mean, I'm a bit intrigued, especially when everyone says that Medtronic is is bad. But yeah, I've just done a quick Google actually. And just sort of seeing what it looks like and what it entails. And obviously I mentioned that I'm really hoping for a closed-loop system to come my way so that's what that's offering already. I'm got very green eyes, because they won't give it to me because I've only just moved to Medtronic so.

P21

Yeah. I think for me, it's definitely something that I would consider. I've sort of been quite sceptical of it just because as we talked about earlier, that sort of lack of control and the element of, you know, giving all the power to the technology in a way. But sort of hearing your guys, positive experiences, experiences of it definitely makes me sort of reconsider my view. And and yeah, I think I'm going to do some more research onto the different closed-loops that are available.

P24

I'm part of a a Facebook group. There is mostly like mums of diabetic kids, but there's a good few diabetic people on there as well, for the t:slim. And I joined that a couple of months before I got it and it just gave me more of like a a real life insight into what it can be like to try to kind of try the ideas out before I actually transitioned onto it.

Interviewer

Yeah. No, that sounds really good was the was there anything that kind of helped in that process of giving up or giving up the control almost or beginning to trust it? Or is there anything actually you think that could be done to make it easier?

P24

Well, they have it set up so that when you start the t:slim, you don't start it with the full closed-loop running, it's like a minimised version of it. So, it'll stop you from going low, but it doesn't give you more insulin if you're going high. So, you can kind of see how you react to that for the first however long, so it's nice to have kind of that stepping stepping stone. It's like a taster and giving up some of the control that won't have any severe effects because you're not having any extra insulin. And then when I was comfortable with that, I was able to move on to the the full control IQ, which will give you more insulin if you're high, which has just been great for me personally.

P19

I I actually didn't. That's interesting said that cause they they threw me in on the full control IQ right from the start. And I actually would have preferred to have done that the restricted version with just the control and the low blood sugars to be honest. But, what helped me was I had a quite a good person from the from the hospital who was and also quite a good rep from Tandem, who genuinely seemed quite knowledgeable and kind of did talk me through some of my somewhat irrational fears. Because, I guess I had a few fears and also some baked in habits, some years of having diabetes, and they just kind of they spent the time talking me through stuff and we did build up slowly. So, at first there wasn't much pressure to kind of get the blood sugars perfect and then overtime we kind of adjusted things. And now I'm in that kind of sweet spot. But yeah, I think it was, it was really good having a couple of people that that were quite knowledgeable about the system already.

P24

I think one thing that like I would have benefited more as having that kind of person who's really knowledgeable about it. I found that the rep from t:slim for me was quite awful to be honest. She wasn't very understanding of what it's actually like to be diabetic and like live like as like just having to manage your pump the whole time. Like I remember a standout moment for me was like, I was commenting on how fiddly the t:slim pump set the set changes are like with the needles that you have to get in the tiny hole and stuff like that. And I just kind of offhand comment was like, “oh, that's gonna be a nightmare when I'm drunk, because sometimes shit happens and you have to do a set change when you're drunk”. And she was like, “well, just don't do it drunk then”. And I'm like. “well, that's not life like like sometimes my pump will run out when I'm drunk, like that's just how it goes”. And I just wish that there was more kind of like eye to eye support in those early days of the pump that [P19], it sounds like you got a much but more understanding rep from Tandem. But yeah, I would have liked to have experience that as well.

Interviewer

No, definitely. And like [P03], what about you? Was there anything kind of if you could go back and tell yourself just before you moved on to the new system, is there anything you'd kind of warn yourself or try and get yourself to do differently?

P03

I. When I first started I had the flexible cannulas and I ended up like getting really really stressed out with them. I think I was just having like a bad week or something and I'd gone for about seven of these flexible cannulas all of them like bending and not working. And I was just like “I can't do this” like and I had a little bit of a cry. So, I think I'd go back and tell myself more about the steel cannulas cause since I've went on the steel cannulas, I've not had any issues like that. And I actually had a similar experience with [P24] where my rep was not a good rep. They were just like, “yeah, this is perfect, everything's perfect” and didn't seem to think about the realities of sometimes you have an occlusion at 2 o’clock in the morning and you can't be bothered to do a full set change. And like they were just like, “well, if you don't know full set change, you're a bad diabetic”. I'm like “it's 2 o’clock in the morning, I'll do it later. But now I'm just putting a new needle in and just saying Bob’s a goodun”. Yeah, they weren't a very understanding of like, day to day life of if you get woken up at 2:00, you're not gonna wanna do like 15 minutes of work when you can do 5.

Interviewer

Yeah, I see. Yeah, I guess I thinking about these companies then and and maybe a little more about the smartwatch side of things if. If those were, I don't know, if Abbott or Tandem or something came out with the smartwatch more designed towards type one diabetes that kind of linked in, is that something people would be interested in or?

P24

I think I would prefer it if if it was the other way, almost. Like if Fitbit or Apple or whatever came out with an app or a collaboration with a closed-loop system. Because they're the they're the people who know smartwatches, I want them to help design a smartwatch and add diabetes tech into it rather than. I think if a diabetes tech company made a smartwatch, it would feel like more medical equipment and just more like another pump that I have to wear essentially.

P03

I agree with that.

P24

But maybe that's just like the, like maybe that's just an association that I've made, I don't know.

P03

No, I I definitely agree with that. I think I'd prefer like Fitbit, Apple, Samsung, Google, whoever makes these smartwatches, to work with Tandem, Abbott, Medtronic, to make something that works. Because there's companies that are making CGMs for fitness and keto and that rubs me the wrong way. I think, in a similar way, Tandem making a smartwatch would rub me in the wrong way cause I'm like that's not what you're aims are. They're putting money that they could. They're spending money that they could put in some more funding for better actual medical equipment into technology that already exists.

Interviewer

Yeah, [P19], it looked like you were, you unmuted to say something. Is there anything you want to add?

P19

No, I I just, I completely agree with with what's already been said. Because I think particularly for me, my my Apple Watch, a very small portion of it or a minority portion of it is diabetes technology. So, I I would never wear one that was primarily geared towards diabetes technology. Just because I don't

like others said I wouldn't trust them to make it good. It wouldn't like for me it wouldn't have the iOS Apple system. It wouldn't have all the stuff that makes me like my Apple Watch or what I bought the Apple Watch. So, actually I think the way round to do it is for the smartwatch makers to make their existing technology more open and available to the developers of the diabetes tech, so they can combine it that way round.

Interviewer

No, definitely there. I mean that all seems to make a lot of sense. Yeah, I guess. Kind of that part of it. And then what other people saying about maybe there are some of the features that aren't so easy to understand. Do you think like if there was this device to come out that had been beginning to integrate type one diabetes technology. Like how important would it actually understanding what it did and how it do it be important to you all?

P06

I think if you pretty important to me.

P24

I think. Sorry my WiFi.

P06

Sorry, I was saying. Yeah, I think it'd be pretty important to me. I think that always I tell myself that it's like it feels a bit uncomfortable not knowing what it's doing so it's like I always tell myself, like I have to know what it's doing. But in reality, I don't know if I'd go through the effort of going through to fully understand what it's doing, so I'd definitely be keen for at least like a surface level explanation of what's going on. Would definitely be good and just give me more more reassurance about the system.

P24

I I second that. I found that like going on to t:slim they, in in their explanation of how it works, it's like “oh, it's a complicated algorithm”. And it's like, although I wouldn't be able to understand much more than that, I would like to be able to have the opportunity of knowing a little bit more than “oh, it's complicated”. And I feel to introduce a new technology, I feel like I would at least want a little bit more than that kind of brushed off explanation.

Interviewer

Yeah. Did you ever manage to get more of an explanation of how the t:slim worked?

P24

Not down to the specifics. They kind of. Like every appointment, I'd kind of get more understanding but it kind of seemed like nobody really knew exactly how it worked. It was more just like, “yep, they have this algorithm. It works.” And that's it.

Interviewer

No, definitely, I mean. Is that something else people agree with, or does anyone think they'd rather just it should be left to get on with their own device? Left to get on, on its own.

P03

No, I I agree. I think with something as important as your life, which is essentially what this technology is, you want to know more of the ins and outs of what it's doing and why it's doing these things. And I think if you know why it makes certain decisions, you can then plan what you're actually doing around that. So, like, for example, sometimes I'll have like a like a litre bottle of Lucozade and drink it up throughout the day. It will like spike me because I've done all my insulin. And then it's like registered it, but then, I've taken too long to drink it so it spikes me and I'm like, “no, I've already done the insulin”. And I think if you know why, where it's making its decisions, you can go, “OK, well, I'm going to do this. This is how the pump will receive that information, here's how I can feel it better so I don't have those spikes and dips.”

Interviewer

Yeah. Is anyone else using a closed-loop kind of felt that, they wanted more options to kind of give the pump more more data or yeah, more information on what they're doing?

P19

Yeah. I I I think I mean for for me, I've always wanted to, especially recently, I want to give my pump more control. I've wanted it to allow it to give me more aggressive corrections because. I know I I I trust it and it's it's proven itself and the way that t:slim was explained to me is it's fixed that Control IQ can't be edited in any way because that's how it's been, you know, regulates and approved and everything. But for me, I'd want to be able, to like you can with your basal rate and your carb rates or whatever, slowly increase and or decrease its kind of involvement over time. Because I actually think for me it would work better if it if it was more aggressive.

Interviewer

Yeah, definitely. I could see that being a thing and kind of trying to have that more control. Yeah, yeah. [P18], how about how would you feel about moving on to a closed-loop if that kind of control element, the fact you’re like giving up general control, but you can't even edit it yourself that easily, if you think something is done wrong?

P18

I definitely feel as though it should action the decision and perhaps send you a notification saying “this is what I'm going to do, you have such and such amount of time to cancel it or override it or adjust it slightly”. Just because, well, we all react insulin differently. So, how is it going to build up an idea or a picture of you when, for example, I'm much more sensitive to insulin than my best friend is because of our statute and like how you know I'm taller, she's much shorter. Like it's even things like that, as well as how I manage stress versus how she manages stress. And unless it can pick up on those very specific profiles I still feel that you should be able to intervene. And then that should be made really accessible, if you've got it on your wrist, for example. And you shouldn't have to go into like a a blood glucose monitor or however you do it. I assume it's the pump then is it, if it's the closed-loop? but that's where there's sort of hub of all the insulin giving things are. But no I just feel like it it doesn't build a proper picture up, but I could be wrong. I guess it would read the the glucose at some stage and think “well that's how they reacted to that”. But then you sort of committed, aren't you? Because if it makes that decision and then you want to go and then go to a water park or something and you wanna take your pump off. I don't know.

Interviewer

Yeah, no, definitely that that makes a lot of sense. Yeah, maybe going back to people saying about, like, understanding what these systems were doing, I guess there's futures where the systems or algorithms within them become increasingly complex. Do you think or how do you think it could be done that this understanding is still conveyed to people who use the systems? Has anyone got any suggestions?

P24

Sorry, could I just pick up on something [P18] said just quickly. I think it one thing you mentioned, [P18], where like the difference in stature of like an insulin sensitivity between you and your best friend, I think it's a real missed opportunity that or like potentially something that could be developed if it's not missed, just too difficult to learn from it mistakes, I guess.

Interviewer

Sorry, [P24], you broke up a little bit for me there, but at the point you were making sound really interesting, do you mind saying that again?

P24

Sorry, I know mine is acting up. I completely died.

Interviewer

Can you hear us all right now, [P24].

P24

Yeah, I can.

Interviewer

Uh, yeah, sorry, I think you broke up a little bit for me in that. Do you mind repeating what you said before?

P24

Yeah. So, I would really like if the closed-loop systems could learn from its mistakes. Like it seems like it would make sense if it could potentially adjust your permanent basals if it recognises that pattern of like, “oh, you're always running a little bit high here for two weeks or however many”. Like I feel like I would be happy to give it that control of adjusting my permanent basal. Or, at least giving me a notification to be like “you've had this pattern, permission to adjust your basal”. Something like that. It just seems like it's a missed opportunity there to not use that data and pattern recognition that it has. Um. And like it would be able to notice say like if if you'd had a high and it corrected and you went really low after it and that happened repeatedly. I don't see any reason why it shouldn't be able to learn that you need less insulin to correct a high and then adjust your insulin sensitivity factors in your pump. That seems like something that could definitely happen in the future.

Interviewer

Yeah, I I think that is that's a really interesting idea. Yeah. Yeah, I guess, [P21], how would you feel about that? Would you trust the system to make those changes?

P21

Yeah, I think I would. One thing I was just thinking, I've just got back from a week’s holiday and for me the the heat, for example, tends to makes me drop lower. And I'm not really sure how it would work with a system like that sort of knowing the temperature. So, for example it might think based on the last week of data that my basal needs to be lowered slightly, but really it's just because I've been in another country and the insulin’s been working faster. So, I think as as some of the others have been saying sort of having that option to override it or say “actually I don't want to reduce my basal permanently or or do so for just you know a week” would would be a good option. So, I'm just thinking I don't know how the pump would know those kind of things, like your location or temperature.

P24

I guess that's where the connection to the smartwatch could even come in, like, cause your your smartwatch has a a skin temperature sensor, and it also has GPS, if you've enabled that. So. I mean theoretically, that there could be a future in which it also takes into account temperature and altitude.

P21

Yeah.

Interviewer

Yeah, definitely. It's really interesting. And there there is more and more information that could go into it. Yeah. [P21], you mentioned on holiday this change that could happen. Is there. What do other people think, might be still points in time, such a system would struggle to manage effectively?

P06

I think like social occasions or exercise, if you're not doing that consistently on the same day for the same amount of time like I would assume it would struggle to keep up with that. Especially, if you are just like meeting up with friends or something and then spur at the moment go off to do something that’s gonna send your blood sugars spiking then it feels like it would be difficult for it to see that coming.

Interviewer

Yeah. So kind of that more things that have an immediate impact. Has anyone else noticed that at all? Yeah. OK, in which case then maybe it is if we go back to what I was saying before and kind of understanding these systems as they become increasingly complicated. Um. Is there is there a way you think kind of it could be conveyed more easily what the system's actually been doing, particularly if it becomes increasingly complicated in terms of the algorithm controlling it at all?

P06

Going off what [P24] said earlier, it gave me the idea of like if it sent you a notification that was making a change. I was wondering if you could have like a summary report at like the end of a week or something. And it can show you like like what, it's what it's done, that it's predicting to do, and then maybe you have to give it permission to activate those changes. And then that way you actually have to like read through what it's doing, so you do understand what it's going to do, maybe not why it's doing it, but maybe that could be also displayed, you just have to go in a bit further.

P24

Yeah, like Dexcom has. Or used to have Dexcom Clarity where you'd be able to look at your averages and it would make basal change suggestions and stuff. It wasn't very good, but this was like a year

ago. So, like having that on an app format, being able to download all the information and really look into all your trends and stuff seems like something that is kind of missing in the market at the moment I guess.

Interviewer

Yeah, definitely. And kind of getting out to check it. Is there anyone who would prefer the more automated side where it just kind of did it for you without checking with you at all?

P18

Think like I mentioned, I would want the smart tech to tell you it's doing something, but then there's a window of you to like interact with it, So, you could get the notification “it's gonna do this, this this”, you think “ohh yeah, that's fine”, put your watch down again and carry on. But then, if you want to do something about it then you would interact with it.

Interviewer

Yeah. No, it's it's interesting. It seems to be a lot of what all the saying is kind of just having that more input yourselves or at least things being checked with you a little more.

P24

Sorry, I just um leading on from I think it was [P18] saying that, that like, I don't want to have to spend more time than I already do thinking about diabetes. So, like being able to just look at notification and be like, “Yep, that's fine” and just put it down seems good. Like I don't, although I want the opportunity, it would be nice to have the opportunity to read through all the information. I don't want that to, I wouldn't want that to have to be a necessity of managing my diabetes. Like the ideal scenario is I have a device that operates as much as like a normal pancreas as possible.

Interviewer

Yeah, I guess that that's interesting then. And so how how do you think? How do you marry those two sides where you do want that input or at least in the initial stages to kind of understand what it's doing versus, yeah, not having to interact with it as much. Do you think there's an easy way around that or any suggestions?

P24

I think it really would just be building that trust with it over time. You know, as as [P19] said, like letting it prove itself. To make it like. I guess that's the only way you'd be able to believe that it works for you, is if it works for you.

P19

Yeah. And I think ultimately the the human needs to have have the final say in the amount of the the level of control and like different people have will be comfortable with different levels of interaction. And so I think the more options it would give, that's then a discussion you can have with your, like your doctor or your pump nurse or whatever based on you, your, your lifestyle, how comfortable you are with the technology. So, for someone like me, I'd give the pump quite high degree of control but then on the other side you could have someone giving the pump a really low degree of control. I think for me it's just giving the end user the ultimate say, so they can make that decision.

Interviewer

No, definitely. I think what you're saying makes me think around this idea that maybe to do the explainability where you ask it more. It's making me think about ChatGPT and kind of, there being that system where you can ask it more questions and it has the option there to kind of explain it to you. Yeah, I wonder how people would feel about that coming from kind of artificial intelligence or a, you know, a computer system rather than necessarily a person or something that has been kind of written up beforehand. I mean, as part of that, has anyone used ChatGPT at all or uses it regularly?

P06

Sorry, I was going to say that I've I've used it a bit and from my understanding it learns off what it's given. So, if it's been like utilised by the company that's written the whole code, then I think I would have quite a high trust in it because it's not going to be giving you misinformation, if it's like the companies told it exactly what it's doing. I think that's quite a good way of saving time so you don't have to call someone else, you can just search it up on your own if you're so inclined. I think it's quite an interesting point, I think [P19] and [P24] were saying earlier about letting it kind of, starting to trust it. I think that's kind of like in my head as not using a closed-loop system at all at the moment, I'm quite hesitant. Like, especially with like the Libre to begin with, I was a bit hesitant, but then like 2 months in, I was totally trusting. I think that like kind of grace period where you start to learn to get used to it, it's quite an important step in making people trust the system. And if you had something like ChatGPT or something that was there to kind of provide support, that might be quite useful.

P03

I think I'd agree with you about the AI, and I think it could go one step further and be like a [P03] AI. So, it's like my specific AI model for my diabetes and it's it can go "ohh well, I did this last time this happened and this was the result. And it didn't work or it worked. So, I can do do something different or the same this time." So, then it almost moulds to your body and how you react to insulin and how all these other factors of you that would come from your wearable would make this make it help it make its decision. And again you could go into into the AI and ask it questions and say "I'm about to go out for a for a run, what do you recommend I do?" Then it can say, "well lower some, we'll lower the background. We'd recommend you take in 15 grammes of carbs before you head out and have 30 grammes afterwards", or some something like that. So then, you're almost having a conversation with your brain and it's got all of your history or the history of all the time you've been using it.

P24

I I think that sounds amazing. Like definitely like like if I was for example, even when, especially when I'm drunk, if I have gone for a night out, I'm going to forget to eat something before I go to bed. But if I had a little thing on my watch, that's like “ohh your blood alcohol content is high, eat something before you go to sleep”. Like it will be brilliant. And I think a lot of people maybe are concerned about the implications of AI having more control control over our health and like the like electronification of health or whatever. But I personally believe that that comes from quite a place of privilege where your body does it for you. It's like, yeah, I would love to not have AI. To need, I would love to not need AI to control my health, but when I have to operate my whole organ myself, actually having AI do that for me would be really nice and the potentially people who oppose that, just don't realise how much their organ does for them anyway.

Interviewer

Yeah, I think that's a really interesting point. I guess there is the other side to it as well, kind of that understandability and explainability side though of, you know learning that trust element. I wonder, like [P18], how do you? I saw you reacting a little earlier, how do you feel about kind of this idea that's been raised about kind of like ChatGPT and having that as kind of a tool that's used as part of the system or something similar based on a technology around it?

P18

As long as in principle, what AI will be doing is building up a a report and sort of studying you based on certain characteristics. What that would entail, I suppose, depends on how clever the model is, like whether you'd have to go through like a full body work up before it can draw a picture of you, or whether you need to like manually input what you think you actually do as a result of exercise or alcohol or anything like that. But it's probably got more an idea of that than any other system because that's how it it learns, isn't it? Like it draws pictures of what it knows to be and then. So, I think it's a a pretty sound model. I I don't know anything really about it, but yeah.

Interviewer

Yeah. I I guess then how would you feel about using that? Would you want to again it to it it to be explained to you quite carefully how it works or?

P18

Not necessarily, because if we're talking, we're going to be able to still see the rationale behind the decision of any sort of diabetes. Whether it's a correction or whether it just thinks or you're going to bed but you, you. Know you've still got an X amount of insulin in your system and you're often not eating enough. Like I I think based on that it's going to know enough to show me it's natural what it wants to do. And then as I said like if it just flashes up on your watch or flashes up on your phone, if you're on your phone. And then you think, “yeah, that's fine”. You don't even have to press like OK, like you just swipe off the notification and if it doesn't hear from you, then that's fine. Because that's how I think you sort of get around not having to think about it too much. Because, if you're deep in a

YouTube video or something, you don't want to have to consciously think too much about it then. I think that's yeah.

Interviewer

Yeah, [P21], what about you? What do you think about this idea that people have suggested around the AI type one diabetes technology?

P21

Yeah, I think it sounds good. I don't know if I'm just a bit sceptical in general, but I feel like I would want to sort of have some sort of training, or some or or know more about it or have like a trial before I fully were to commit to trusting it or or learning it, manage my diabetes completely. Um. But I think some of the suggestions have been have been really good.

Interviewer

Yeah. Is there anything particular you'd want from that training or them to do?

P21

I guess maybe just like an explanation of how how it's going to make those decisions and. I don't know. I don't know if like how you can maybe override them or I don't know if that if that's an option. But yeah, I think just for me, like I I do quite like to understand why things are the way they are and the sort of reasoning and rationale behind them. Um. So, I don't think I would necessarily trust a system just to work straight away. Though I'm sure like obviously to get approved, these things need to be, you know, tested loads, but I would just want to have that sort of training, I think. Or, yeah, even like case studies or or something where people, it shows that it's it's work for people. Even if that just involves speaking to someone that's used it.

Interviewer

Right. No, that makes a lot of sense. Um. Yeah. Can anyone see any potential risks though to that system or times where it might go wrong?

P19

I think with like with AI has been, you know, I I read something the other day where the the kind of the efficacy of ChatGPT has gone down significantly as it's hit the mainstream cause it's. I don't. I don't claim to know anything at all about how ChatGPT works, but apparently, it's getting worse. And I I tried this out in uni, I asked it to find me some information and asked to cite its sources and it started coming up with completely made-up sources and so. But it was very convincing, it thought it was right. So, I just thought I think if that's applied to a diabetes model you, ultimately I'd wanna know what the limits are. So, is it is this thing gonna learn from other users and would those other

users potentially negatively impact you or is it only gonna learn from you? Cause only gonna learn from you, it's probably not gonna go wrong, but if it's kind of connected to a wider you know language learning model, it could go wrong a little bit wrong. Like how ChatGPT is going a little bit wrong at the moment.

P24

Yeah, I would definitely be concerned if it if my my diabetes was essentially also being influenced by other people's diabetes because everyone responds differently to different stimuli. I would want it to be a, I would want it to be an only me kind of learning situation.

P03

I've done a quick Google and it's for Type 2, unfortunately, but in America they're testing an AI model which has had an 84% success rate in reversing type 2 diabetes, of like whole body digital twin, which is what the AI is called. So, it is just [P03] AI, so it's just your twin. It's your AI model. So, it would be individual to you and learning off of you. And I think.

P24

That sounds great.

P03

Yeah, I mean it's for type 2 diabetes at the minute, but there's hope. That means I've not, I've not read the article and I don't think they've mentioned type one diabetes, but there's hope if they're doing something like that for Type 2 that something like that for type one could be coming. If you're interested in reading about it is called whole body digital twin.

Interviewer

Yeah. What do people think about that? Is that something they'd be interested or like the sound of?

P24

I would definitely do that. Even just like. I'm one of those people that I really am not that bothered about my personal data and and like any protection of that. Like I've done 23andMe and I've learned all my genetics and like I like I've really enjoyed having a Fitbit and that kind of thing. And just having that personalised approach, it really appeals to me and my health management.

Interviewer

Yeah. Anyone else at all? That all makes a lot of sense.

P06

Yeah, I think I'd be pretty up for trying it. I'm pretty happy to try any. I think I I tell myself, I'm pretty happy to try new things to see if it actually helps with diabetes. I'm sure if it then messed up, I'd be quite upset, but no, I think I'd be willing to give it a go.

Interviewer

I think that's quite an interesting idea though that, yeah, if it does mess up, I mean, how do you how does everyone think they'd react if they did start using this system and it wasn't quite as effective as what they're currently doing? Would they keep using it, or do you kind of then tend to scrap that idea for now and just go back to what you're doing?

P06

I'd probably scrap it until it's then got like recorded good results. Yeah, until there's like a bit of trust built up in it.

Interviewer

And what would count as that trust for you?

P06

Success stories, I guess. Or just like data of other people having good experiences, or if the issue I did have been removed. That kind of thing.

P24

I would say that I had kind of that thought sort of process with the t:slim like with the Facebook group that I joined beforehand. It was very much a case of I want to see this working for people. I'd heard about it being great in the US, but quite frankly, I don't really like how much value to the US medical opinions. So, after it came to the UK, I still kind of gave it a bit of time to see how people in the UK were responding to having it on the NHS and how the prescriptions went. And, you know, just made sure that everybody was having a good experience and there were success stories before. I considered it for myself.

P06

Yeah, I think if there was something like that, I definitely feel much more secure in making a decision.

Interviewer

I think that's another interesting point [P24] makes around kind of the, yeah, these different shifts from different areas and in the US, there's quite a lot of stuff that's trialled before or more heavily worked. Um. How important would the rest of you say are kind of seeing results that are UK-based to you?

P03

I I would say having UK based results is important to me. I think every country, especially the US, has different like food regulations, which is. So like a slice of bread in the America is very different, like a slice of bread here. And I'd want to know based on like what people have experienced here with our, with the weird British microclimate. I'd want to know like all those specifics because they're all factors and I think they're what's important to me. It'd be great to hear American success stories, but you don't know if that's because people have been able to buy top of the range stuff and then it's come here and it's not on the NHS, so you're you'd have to pay for it. It's the same with other countries, Australia, they have their own thing. All the European countries, Asian countries, they're all so individual, I think you'd want to have success within your country before you venture out on trust a new piece of technology.

Interviewer

Yeah, that that makes a lot of sense to me. I guess the final the final thing I really wanted to talk about then, which is come up a little bit already, yeah, the times where these things go wrong, I guess initial stages. I mean people were mentioning the the Libre app recently. What, what do you? So yeah, if we take this the Libre app breaking, which I I guess for I mean [P21] and [P06], you said you kind of experienced this more first hand, but for the rest of you, yeah, there was the app that kind of on iPhones. And I think it was more UK focused than in other areas as well. It stopped working for a while and then if you deleted it, it wasn't available to redownload that took them kind of half a day or more to really sort this issue out before people could download it. What do you think should be done in these situations to kind of better manage them or better protect people using them?

P24

I mean, I feel like the brands need to be really responsive to their customers, like purely in a customer and producer kind of relationship. Like it's a different thing but I remember back in January last year, Dexcom changed their type of adhesive without informing people. And I was one of the people who started getting really bad allergic reactions to the new adhesive that they used. Um. And it took them like two weeks to first update anyone about the adhesive changes and another two months or something like that to make further changes so that fewer people would have reactions again. Um. And I just think like Libre like taking about half a day sounds reasonable I think for something that clearly wasn't an unexpected issue that they had. Obviously, it's stressful in the moment, especially if you had something big going on that morning that you needed to know your bloods for, I think. These companies need to be very aware that any changes that they make are going to impact our everyday lives and every single minute of them. It's not just like, oh, we tried a new flavour of something, it's it's way more serious than that.

Interviewer

Yeah. How how about you, [P18], what do you think about that?

P18

Yeah, I I believe. I think same things really because if an app's going to go down or it's going to crash, then it will probably do so quite magnificently. Obviously, I'm experiencing some issues as it is and it is quite inconvenient, but that's also taught me to keep my blood glucose metre quite close by. And I know it's it's difficult because you do want to just get in your life and get into the technology side of things and not carry it around with you. But I need it for my job, so I need to test my blood before I drive, otherwise I'll take my licence off me. So, it is always there like the glucose machines. So I feel. like in the grand scheme of things, if every diabetic is carrying a blood glucose machine with them, then it wouldn't be that bad. But, subsequently, the expectation shouldn't have to be that in this day and age you have to be able to take a a glucose machine with you to draw blood all the time. But then I've lost my phone before now and had no means of well doing a test. And then it's the same with my actual insulin. Like I don't carry a pen beyond my pump. So then, I've been somewhere before where I've, like, ripped my cannula out and I've then had no means to do that either. So, and while, some little blip in the system or blip in like an app could be quite annoying and lengthy. If it's going to do it, it's going to do it, isn't it? And you just, you either have the stuff or you have to feel a bit shit for a while. Sorry for my language.

Interviewer

No, it's no worry. Not I think you make a very good point as well, but yeah it doesn't matter what system you use, these things can't go wrong, even if it is more advanced. Yeah, I guess, what, what are people's reactions in those times where it does go wrong? What what's the first thing you do? I mean [P19] say, have you ever kind of been in position where the tech you're using hasn't worked out so well? What what's kind of been your reaction in these times?

P19

Yeah, I mean it, it depends what's happened. Um. I'm still pretty good at carrying a blood glucose machine because obviously you need it when you're driving. Um. But there are times when I don't, when I don't carry it. It it does, it does depend really on kind of how how severe it was, I mean, this morning when the insulin pump shut down, I was seriously considering how to get the old Medtronic one out and start up going over to the Medtronic again because I didn't know if it was gonna turn back on again. I think if I'm away, if I'm away from home just for a day or whatever and say I haven't bought an insulin pen. I would probably if I say like say, pull the cannula out I just have to kind of stuck it up for for the rest of that day until I could get home and and change my cannula if I hadn't bought a spare set with me. But, I do try and have spares like in my bags, in my backpacks, in the car and everything, so hopefully I'm never too far away from a kind of a replacement unit.

P24

Yeah, I try to do the same thing. I find that the main time where I've found myself in a in a pinch is when I'm going out for the night, like going clubbing. I don't want to have a big bag on me, so I'm trying to minimise as much as possible. I've actually done set changes in club bathrooms, which probably isn't the most sanitary thing, but shit happens. And I I've had to say to the like the person that stands in there selling your perfumes like, “sorry, I'm not about to do heroin, I'm diabetic, but I'm going to have a needle. Don't panic.” It's quite funny. But yeah, like, you're not gonna remember every time. And I've had to go into ambulances and be like, “hi. I'm fine but can I just check my bloods, my Dexcom’s just run out”. And like just use all the resources that are available, so that I don't have to make an extra trip home and back, or even to the hospital if it gets that bad.

Interviewer

Right. I see. Yeah. Can people. I mean, do you think that's something you're always gonna have to do? There's always gonna have to be stuff you carry with you or can anyone see a kind of a future where that isn't needed?

P24

I mean, whenever I talk to someone who's just not diabetic then, like, “why can't you have a pump that also gives you glucose?” And I've never been able to give them an answer. I I know there must be one because it seems too obvious, but like that does seem like something that could happen one day, where there's some sort of way that your pump can bring your blood up, instead of just stopping them from going as low.

P03

I mean, I've I've had that question to myself. Why can't I have a pump that gives me glucose? I think the the most logical answer I've ever come up with for myself is that like glucose is a lot thicker than insulin. But then again, you can get glucose on an IV if you're in hospital, so they can figure they.

P24

So they should figure it out.

P03

They can figure it out.

P18

There are hypo stop injections as well, the orange.

P03

The glucagon.

P18

Injections, yeah.

P03

They're not glucose, though glucagon is a different hormone.

P18

Either way, you can put it in a blooming cartridge and put it in a pump, in my opinion, yeah.

P24

Yeah, they should have figured out by now. Yeah, it does seem like there will be a future where you don't have to carry around 4 packs of Haribo Minis and a gluco gel and know where the nearest ambulance is, just in case you get a vodka Diet Coke instead of a vodka Coke.

Interviewer

Yeah, no, definitely.